

NORMAL INCIDENCE (boundary at z = 0)

Medium 1 (η_1, k_1) for $z < 0$; Medium 2 (η_2, k_2) for $z \geq 0$.
Reflection & transmission coefficients

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}, \quad \tau = \frac{2\eta_2}{\eta_2 + \eta_1} = 1 + \Gamma$$

Power conservation

$$|\Gamma|^2 + \frac{\eta_1}{\eta_2} |\tau|^2 = 1$$

Fields in medium 1 *incident + reflected*

$$\tilde{\mathbf{E}}_1 = E_0^+ (e^{-jk_1 z} + \Gamma e^{+jk_1 z}) \hat{x}$$

$$\tilde{\mathbf{H}}_1 = \frac{E_0^+}{\eta_1} (e^{-jk_1 z} - \Gamma e^{+jk_1 z}) \hat{y} \quad (\text{note } -\Gamma)$$

Fields in medium 2 *transmitted only*

$$\tilde{\mathbf{E}}_2 = \tau E_0^+ e^{-jk_2 z} \hat{x}, \quad \tilde{\mathbf{H}}_2 = \frac{\tau E_0^+}{\eta_2} e^{-jk_2 z} \hat{y}$$

Special cases			
Case	η_2	Γ	τ
PEC (perfect cond.)	0	-1	0
PMC	∞	+1	2
Matched	η_1	0	1

STANDING WAVES & VSWR

VSWR (voltage standing-wave ratio)

$$S = \frac{1 + |\Gamma|}{1 - |\Gamma|} \in [1, \infty) \Leftrightarrow |\Gamma| = \frac{S - 1}{S + 1}$$

Envelope in medium 1

$$|\tilde{E}_1|_{\max} = |E_0^+|(1 + |\Gamma|), \quad |\tilde{E}_1|_{\min} = |E_0^+|(1 - |\Gamma|)$$

PEC ($\Gamma = -1$) standing wave — zero $\hat{k}$ component

$$|\tilde{\mathbf{E}}_1(z)| = 2|E_0^+| |\sin(k_1 z)|$$

Zeros at $z = 0, -\lambda_1/2, -\lambda_1, \dots$ (multiples of $\lambda/2$)
Maxima at $z = -\lambda_1/4, -3\lambda_1/4, \dots$

Net power flow in medium 1

$$S_{1,\text{av}} = \frac{|E_0^+|^2}{2\eta_1} (1 - |\Gamma|^2) \hat{z}$$

PEC: zero net power ($|\Gamma| = 1$).

OBLIQUE INCIDENCE

θ_i = incidence angle (from normal); $\theta_r = \theta_i$;
 θ_t = transmitted.

Snell's law $n = \sqrt{\mu_r \epsilon_r}$

$$n_1 \sin \theta_i = n_2 \sin \theta_t, \quad n = \frac{c}{u_p} = \sqrt{\mu_r \epsilon_r}$$

Non-magnetic ($\mu_{r1} = \mu_{r2} = 1$): $\sqrt{\epsilon_1} \sin \theta_i = \sqrt{\epsilon_2} \sin \theta_t$.
TE polarization **E** $\perp$ plane of incidence ("s")

$$\Gamma_{\text{TE}} = \frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$$

$$\tau_{\text{TE}} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$$

TM polarization **H** $\perp$ plane of incidence ("p")

$$\Gamma_{\text{TM}} = \frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$$

$$\tau_{\text{TM}} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$$

Power reflectance & transmittance

$$R = |\Gamma|^2, \quad T = 1 - R = \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i} |\tau|^2$$

BREWSTER'S & CRITICAL ANGLES

Brewster's angle $\Gamma_{\text{TM}} = 0$; non-magnetic only

$$\tan \theta_B = \frac{n_2}{n_1} = \sqrt{\frac{\epsilon_2}{\epsilon_1}} \quad (\text{no TE Brewster's})$$

At θ_B TM polarization is **fully transmitted**; TE still partly reflected.

Critical angle *TIR*; requires $n_1 > n_2$

$$\sin \theta_c = \frac{n_2}{n_1}, \quad \theta_c = \arcsin\left(\frac{n_2}{n_1}\right)$$

For $\theta_i \geq \theta_c$: $|\Gamma| = 1$ (both TE and TM), evanescent wave in medium 2.

Optical fiber (step-index) $n_{\text{core}} > n_{\text{clad}}$

$$\text{NA} = n_0 \sin \theta_a = \sqrt{n_{\text{core}}^2 - n_{\text{clad}}^2}$$

θ_a = acceptance half-angle (outside). $n_0 = 1$ in air.

WAVE IMPEDANCE & MULTI-LAYER

Input impedance at $z = -d$ looking toward load Z_L at $z = 0$, medium η :

$$Z_{\text{in}} = \eta \frac{Z_L + j\eta \tan(\beta d)}{\eta + jZ_L \tan(\beta d)}$$

Quarter-wave transformer ($d = \lambda/4$, $\tan \rightarrow \infty$)

$$Z_{\text{in}} = \frac{\eta^2}{Z_L} \Rightarrow \eta = \sqrt{Z_{\text{in}} \cdot Z_L} \text{ for match}$$

Half-wave ($d = \lambda/2$) *transparent*

$$Z_{\text{in}} = Z_L$$

PEC load ($Z_L = 0$) *short circuit*

$$Z_{\text{in}} = j\eta \tan(\beta d) \quad (\text{pure reactance})$$

3-layer anti-reflection coating $\lambda/4$ slab, medium η_2

$$\Gamma = 0 \Leftrightarrow \eta_2 = \sqrt{\eta_1 \eta_3}, \quad d = \lambda_2/4$$

REFLECTION QUICK TABLE

$ \Gamma $	VSWR	% reflected	% transmitted
0	1.0	0%	100%
0.33	2.0	11%	89%
0.5	3.0	25%	75%
0.75	7.0	56%	44%
1.0	∞	100%	0%

USEFUL TRIG FOR OBLIQUE

Find $\cos \theta_t$ after Snell's:

$$\cos \theta_t = \sqrt{1 - \sin^2 \theta_t} = \sqrt{1 - \frac{n_1^2}{n_2^2} \sin^2 \theta_i}$$

Complex if $\theta_i > \theta_c$ (TIR — $\cos \theta_t$ becomes imaginary).
Check: for normal incidence ($\theta_i = 0$): $\cos \theta_i = \cos \theta_t = 1$;
Fresnel $\rightarrow$ *normal-incidence* Γ .

Γ/τ QUICK

$$\Gamma = (\eta_2 - \eta_1)/(\eta_2 + \eta_1)$$
$$\tau = 1 + \Gamma$$
$$|\Gamma|^2 + (\eta_1/\eta_2)|\tau|^2 = 1$$
$$S = (1 + |\Gamma|)/(1 - |\Gamma|)$$

PEC STANDING

$$\Gamma = -1; \tau = 0$$
$$|\tilde{E}| = 2|E_0| |\sin(kz)|$$

Zeros: $z = 0, -\lambda/2, \dots$
Max: $z = -\lambda/4, -3\lambda/4, \dots$

SNELL'S / ANGLES

$$n_1 \sin \theta_i = n_2 \sin \theta_t$$
$$\sin \theta_c = n_2/n_1 \text{ (TIR)}$$
$$\tan \theta_B = n_2/n_1 \text{ (TM)}$$
$$\text{NA} = \sqrt{n_c^2 - n_{\text{cl}}^2}$$

$\lambda/4$ TRANSFORM

$$Z_{\text{in}} = \eta^2/Z_L$$

Match: $\eta = \sqrt{Z_1 Z_3}$
 $\lambda/2$: $Z_{\text{in}} = Z_L$
AR coat: $\eta_2 = \sqrt{\eta_1 \eta_3}$

FRESNEL QUICK

TE: $\Gamma = (\eta_2 c_i - \eta_1 c_t)/(\dots)$
TM: $\Gamma = (\eta_2 c_t - \eta_1 c_i)/(\dots)$
 $c_i = \cos \theta_i$; $c_t = \cos \theta_t$
 $R = |\Gamma|^2$; $T = 1 - R$

RADIATION FUNDAMENTALS

Far field ($r \gg \lambda$): only $\hat{\theta}$ and $\hat{\phi}$ components survive. $\hat{r}$ component decays as $1/r^2$.
Radiation intensity W/sr ; r^2 times power density

$$U(\theta, \phi) = \frac{r^2}{2\eta_0} |\vec{E}_\theta|^2 \text{ (W/sr)}$$

Total radiated power

$$P_{\text{rad}} = \iint_{\Omega} U \, d\Omega = \int_0^{2\pi} \int_0^\pi U \sin \theta \, d\theta \, d\phi$$

Normalized pattern $F_{\text{max}} = 1$

$$F(\theta, \phi) = \frac{U(\theta, \phi)}{U_{\text{max}}}$$

HERTZIAN (SHORT) DIPOLE

Length $dl \ll \lambda$, current I_0 , at origin, oriented $\hat{z}$:
Far-field phasors

$$\vec{E}_\theta = j \frac{\eta_0 I_0 \, dl \sin \theta}{2\lambda \, r} e^{-jkr}, \quad \vec{H}_\phi = \frac{\vec{E}_\theta}{\eta_0}$$

Radiation intensity

$$U(\theta) = \frac{\eta_0}{8} \left(\frac{I_0 \, dl}{\lambda} \right)^2 \sin^2 \theta, \quad F(\theta) = \sin^2 \theta$$

Total radiated power

$$P_{\text{rad}} = \frac{\pi}{3} \eta_0 \left(\frac{I_0 \, dl}{\lambda} \right)^2 = 40\pi^2 |I_0|^2 \left(\frac{dl}{\lambda} \right)^2$$

Radiation resistance

$$R_{\text{rad}} = \frac{2P_{\text{rad}}}{|I_0|^2} = 80\pi^2 \left(\frac{dl}{\lambda} \right)^2 \, \Omega$$

Directivity: $D_0 = 1.5$ (1.76 dBi); HPBW= 90.

HALF-WAVE DIPOLE ($\ell = \lambda/2$)

$$F(\theta) = \left[\frac{\cos(\frac{\pi}{2} \cos \theta)}{\sin \theta} \right]^2$$

$$R_{\text{rad}} = 73 \, \Omega, \quad D_0 = 1.64 \text{ (2.15 dBi)}, \quad \text{HPBW} = 78$$

$Z_{\text{in}} = 73 + j42.5 \, \Omega$ (at $f = c/2\ell$; slightly shorter for resonant)

ANTENNA PARAMETERS

Directivity *dimensionless*

$$D_0 = \frac{4\pi U_{\text{max}}}{P_{\text{rad}}}$$

Radiation efficiency & gain

$$e_{cd} = \frac{P_{\text{rad}}}{P_{\text{rad}} + P_{\text{loss}}} \in [0, 1], \quad G_0 = e_{cd} D_0$$

Effective aperture m^2

$$A_{\text{eff}} = \frac{\lambda^2 G_0}{4\pi}$$

Beam solid angle

$$\Omega_A = \iint_{\Omega} F \, d\Omega = \frac{4\pi}{D_0} \text{ (sr)}$$

HPBW (half-power beamwidth) *angle between $F = 0.5$ pts*

$$D_0 \approx \frac{4\pi}{\theta_{1\text{HP}} \theta_{2\text{HP}}} \text{ (approx, narrow beam)}$$

In dB $G_{\text{dBi}} = 10 \log_{10}(G_0)$; isotropic ref

$$G_{\text{dBi}} = 10 \log_{10}(G_0), \quad 0 \text{ dBi} \Rightarrow G_0 = 1$$

FRIIS TRANSMISSION

Free space, polarization matched, separation R :

$$\frac{P_r}{P_t} = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2$$

In dB form

$$P_r[\text{dBW}] = P_t[\text{dBW}] + G_t[\text{dB}] + G_r[\text{dB}] - \text{FSPL}[\text{dB}]$$

$$\text{FSPL} = 20 \log_{10} \left(\frac{4\pi R}{\lambda} \right) \text{ dB}$$

Equiv.: $P_r = S_i A_{\text{eff},r}$ where $S_i = P_t G_t / (4\pi R^2)$.

RADAR EQUATION

Monostatic radar $G_t = G_r = G$, range R

$$\frac{P_r}{P_t} = \frac{G^2 \lambda^2 \sigma_{cs}}{(4\pi)^3 R^4}$$

Maximum range

$$R_{\text{max}} = \left[\frac{P_t G^2 \lambda^2 \sigma_{cs}}{(4\pi)^3 P_{r,\text{min}}} \right]^{1/4}$$

σ_{cs} =radar cross section (m^2); $P_{r,\text{min}}$ =minimum detectable power.
Power falls as $1/R^4$: doubling range requires $\times 16$ transmit power.

RECTANGULAR WAVEGUIDE ($a \times b$, $a > b$)

Propagation in $+\hat{z}$. Modes: TE_{mn} ($E_z = 0$) or TM_{mn} ($H_z = 0$).
Cutoff wavenumber & frequency

$$k_{c,mn} = \pi \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}, \quad f_{c,mn} = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

TE_{mn} : $m, n = 0, 1, 2, \dots$ (not both 0). TM_{mn} : $m, n \geq 1$.
Dominant mode TE_{10} ($m=1, n=0$): $f_c = c/(2a)$ (lowest).
Single-mode range *only TE_{10} propagates*

$$\frac{c}{2a} < f < \frac{c}{a} \quad (\text{i.e. } f_{c,10} < f < f_{c,20})$$

WAVEGUIDE WAVE PARAMETERS ($f > f_c$)

Propagation constant

$$\beta = \sqrt{k_0^2 - k_c^2} = \frac{\omega}{c} \sqrt{1 - \left(\frac{f_c}{f}\right)^2} \text{ (real, } f > f_c)$$

Phase velocity ($> c$) and group velocity ($< c$)

$$u_p = \frac{\omega}{\beta} = \frac{c}{\sqrt{1 - (f_c/f)^2}} > c, \quad u_g = \frac{c^2}{u_p} = c \sqrt{1 - \left(\frac{f_c}{f}\right)^2} < c$$

$$\mathbf{u}_p \cdot \mathbf{u}_g = c^2$$

Guide wavelength

$$\lambda_g = \frac{2\pi}{\beta} = \frac{\lambda_0}{\sqrt{1 - (f_c/f)^2}} > \lambda_0$$

Wave impedances

$$\eta_{TE} = \frac{\eta_0}{\sqrt{1 - (f_c/f)^2}} > \eta_0, \quad \eta_{TM} = \eta_0 \sqrt{1 - \left(\frac{f_c}{f}\right)^2} < \eta_0$$

TE₁₀ FIELDS

Amplitude H_0 ; x -dependence $\sin(\pi x/a)$:

$$\vec{H}_z = H_0 \cos\left(\frac{\pi x}{a}\right) e^{-j\beta z}, \quad \vec{H}_y = 0$$

$$\vec{E}_y = -\frac{j\omega\mu a}{\pi} H_0 \sin\left(\frac{\pi x}{a}\right) e^{-j\beta z}$$

$$\vec{H}_x = \frac{j\beta a}{\pi} H_0 \sin\left(\frac{\pi x}{a}\right) e^{-j\beta z}$$

$$\vec{E}_x = \vec{E}_z = \vec{H}_y = 0$$

Only $\hat{y}$ component of $\mathbf{E}$; vanishes at $x = 0$ and $x = a$.

DIPOLE QUICK

Hertzian: $R_{\text{rad}} = 80\pi^2 (dl/\lambda)^2$
 $D_0 = 1.5$; HPBW= 90
Half-wave: $R_{\text{rad}} = 73 \, \Omega$
 $D_0 = 1.64$; HPBW= 78

ANTENNA PARAMS

$G_0 = e_{cd} D_0$
 $A_{\text{eff}} = \lambda^2 G/(4\pi)$
 $\Omega_A = 4\pi/D_0$
 $D_0 \approx 4\pi/(\theta_1 \theta_2)$

FRIIS / RADAR

$P_r/P_t = G_t G_r (\lambda/4\pi R)^2$
FSPL= $20 \log(4\pi R/\lambda)$
Radar: $P_r/P_t =$ $=$
 $G^2 \lambda^2 \sigma / ((4\pi)^3 R^4)$
 $R_{\text{max}} \propto (P_t G^2 \sigma)^{1/4}$

WAVEGUIDE

$f_c = c/(2a)$ [TE₁₀]
 $\beta = k_0 \sqrt{1 - (f_c/f)^2}$
 $u_p u_g = c^2$
 $\eta_{TE} = \eta_0 / \sqrt{1 - (f_c/f)^2}$

TE₁₀ QUICK

$\vec{E}_y \propto \sin(\pi x/a)$
 $\vec{H}_z \propto \cos(\pi x/a)$
 $\vec{H}_x \propto \sin(\pi x/a)$
 $\lambda_g = \lambda_0 / \sqrt{1 - (f_c/f)^2}$